

Entropy-Driven Sampling for Remote Estimation in Internet of Things Systems

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Abstract—Remote state estimation plays a critical role in Internet of Things (IoT) systems, where distributed sensors report dynamic processes to remote estimators for control and monitoring. Traditional freshness metrics, such as age of information (AoI), do not necessarily capture the information value degradation caused by random transmission latency and content dynamics. In this article, we evaluate the value of information (VoI) through the lens of Shannon entropy by adopting the *uncertainty of information* (UoI), which quantifies the receiver's uncertainty in estimating the source state caused by stale information delivery. We study a remote estimation system where a binary Markov source is sampled and transmitted over a randomly delayed channel with a long-term sampling frequency constraint. This problem is formulated as a constrained partially observable semi-Markov decision process (C-POSMDP), solved through Dinkelbach's transformation and Lagrangian relaxation combined with the relative value iteration (RVI) algorithm. To mitigate the complexity of RVI, we further develop a low-complexity index-based policy by approximating the Bellman equation under relatively high delay. Numerical results show that both proposed policies outperform zero-wait, uniform, and AoI-optimal baselines, with the suboptimal policy achieving near-optimal performance in relatively high delay regimes.

Index Terms—Age of information (AoI), entropy, Markov decision process (MDP), remote state estimation, uncertainty-of-information (UoI).

I. INTRODUCTION

ENSURING the timeliness of information delivery has become an essential requirement in the Internet of Things (IoT), particularly for remote monitoring and estimation applications such as industrial automation, vehicular networks, and uncrewed aerial vehicle (UAV)-assisted edge systems. In these scenarios, distributed sensors continuously monitor dynamic physical processes and transmit status updates to

remote estimators or controllers through wireless channels. To evaluate the timeliness of these updates, the age of information (AoI) has emerged as an important performance metric since its introduction in [2] and [3]. AoI has been widely applied in designing update policies for remote estimation and monitoring systems [4], [5], [6], in analyzing and optimizing queuing and scheduling strategies [7], and in designing communication and network protocols [8], [9]; see [10] for a comprehensive survey. However, the classical AoI metric is inherently *state-agnostic*, as it only captures the time elapsed since the last received update, without accounting for the source dynamics or semantic importance of the data [11]. To address this limitation, recent research efforts have proposed generalized AoI-inspired metrics tailored to diverse applications or analytical scenarios. These metrics can be broadly categorized into three types: 1) monotonic penalty functions of AoI [12], [13]; 2) nonmonotonic functions of AoI [14], [15], [16], including L-conditional entropy; and 3) semantics-aware metrics [17], [18], [19], [20], such as age of incorrect information (AoII).

A. Motivations

Despite the aforementioned development of AoI variants, a notable question remains relatively unaddressed: *under a remote estimation scenario, how stale observations impact the receiver's knowledge of the current source state*. This motivates an information-theoretic perspective to quantify this impact [21], [22], [23], [24], [25], [26]. Among these works, Sun and Cyr [21] proposed using Shannon mutual information between the real-time source state and the received samples to quantify this impact. Notably, it considered the estimation of a special *symmetric* binary Markov source, which aligns closely with our setup, allowing the mutual information to be expressed as a nonincreasing function of AoI. A closely related information-theoretic metric, termed value of information (VoI), was introduced in [24] for hidden Markov models, where information freshness is quantified by the mutual information between the current state and the observation history. In this setting, VoI is shown to be a monotonic function of AoI and does not depend on the specific value of the most recent sample.

However, for a more general case where the Markov source is **not symmetric**, the relevant information-theoretic penalty can no longer be expressed as an explicit, monotonic function with respect to AoI. To quantify the knowledge loss of information due to staleness in such asymmetric

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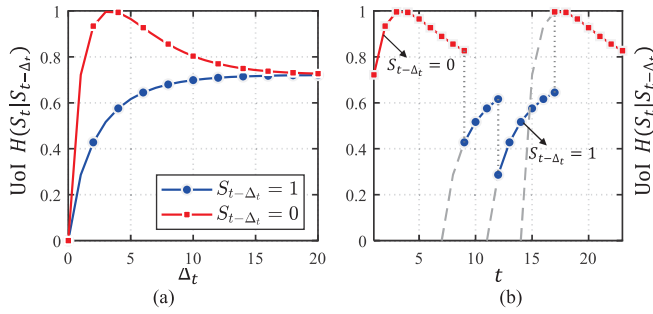


Fig. 1. UoI evolution under an asymmetric binary Markov source, where $\Pr\{S_{t+1} = 0|S_t = 1\} = 0.2$ and $\Pr\{S_{t+1} = 1|S_t = 0\} = 0.05$. (a) UoI versus AoI. (b) UoI over time.

cases, the uncertainty of information (UoI) has been proposed [22], [23]. UoI is defined as the conditional entropy of the current source state given the receiver's past observations. This metric is particularly pertinent to asymmetric monitoring applications, such as industrial anomaly detection or forest fire alerting. In such systems, the source predominantly stays in a “normal” state (e.g., State 0) and rarely transitions to an “abnormal” state (e.g., State 1). UoI captures the nuanced reality that the growth of uncertainty—and thus the urgency to sample—depends on whether the last observation was “normal” or “abnormal.” This state-dependent and nonuniform characterization of staleness-induced uncertainty, while more expressive, alters the structural properties of the performance metric. Specifically, the following holds.

- 1) *Nonmonotonicity*: Unlike mutual information under symmetric sources, UoI is a nonmonotonic function in terms of AoI, which depends on the transition probability matrix of the source. As shown in Fig. 1(a), UoI may initially increase and then decrease with rising AoI.
- 2) *State Dependence*: The evolution of UoI is conditioned on the last observed state $S_{t-\Delta(t)}$, where S_t is the current ground-true source state and Δ_t denotes the instantaneous AoI at time slot t . This introduces inherently *state-aware* and *content-sensitive* dynamics. Fig. 1(b) shows that UoI evolves differently depending on whether the last received update was $S_{t-\Delta_t} = 0$ or 1, leading to diverging update trajectories over time.

These properties make the design of optimal sampling policies more complex. Existing approaches have focused on an *ideal one-time-slot* delayed framework [22], [23], in which a binary decision—whether to sample and transmit—is made at each time step. Under this setup, decisions are made with perfect foresight, and channels incur constant delays.

Such assumptions are unrealistic in remote estimation for IoT systems, where random latency, retransmissions, and congestion are common, and where energy or sampling frequency constraints further limit update opportunities. In such systems, the one-time-slot delay assumption adopted in existing UoI-optimal policies no longer holds, leading to degraded performance when updates experience random and time-varying delays. To address this gap, it is essential to develop a new class of sampling policies that simultaneously consider random delays and the constraints of resource-limited environments.

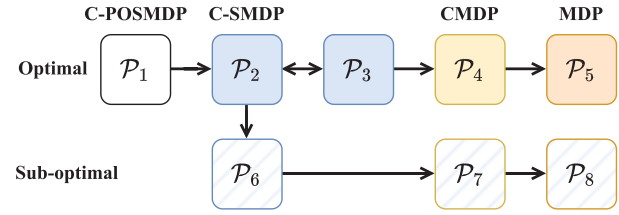


Fig. 2. Relationship between formulations.

B. Contributions and Novelty

Our main contributions are summarized as follows.

- 1) *Problem Formulation*: This work is a direct extension of [1], [22], [23], which integrates random transmission delays and sampling frequency constraints into the UoI framework. These practical considerations fundamentally reshape the problem, where *per-slot* binary transmission decisions no longer suffice, and the conventional infinite-horizon Markov decision process (MDP) formulation becomes inapplicable. To address these challenges, we propose a novel formulation that models the UoI-minimization sampling problem as a constrained partially observable semi-Markov decision process (C-POSMDP) [27]. This new formulation allows us to jointly handle dynamic uncertainty, random transmission delay, and sampling frequency constraints. *To the best of our knowledge, this is the first work to employ a C-POSMDP formulation for optimal freshness-driven sampling design.*
- 2) *Optimal Sampling*: We optimally solve the formulated C-POSMDP through a *divide-then-concur* approach, as shown in Fig. 2. First, we introduce the belief state, which serves as a sufficient statistic for the observation history, allowing the C-POSMDP to be reformulated as a constrained semi-Markov decision process (C-SMDP). Then, we apply Dinkelbach's method [28] for fractional programming, which transforms the C-SMDP into an infinite-horizon constrained Markov decision process (CMDP) under a given Dinkelbach parameter. Finally, we solve this CMDP using a Lagrangian relaxation approach. To compute the optimal policy, we design a three-stage algorithm that involves bisection search and *relative value iteration* (RVI). We further provide a rigorous convergence analysis, proving that the RVI procedure is convergent in our system setup.
- 3) *Low-Complexity Index Sampling*: The computational cost of solving the CMDP via the RVI algorithm motivates the development of a low-complexity, index sampling policy. However, designing such a policy is *nontrivial* due to two intrinsic properties of UoI: its **nonmonotonicity** renders fixed thresholds ineffective, and its **state dependence** requires the threshold to adapt dynamically to varying observation contexts. To overcome these challenges, we conduct an approximate analysis of Bellman's equation, which reveals a structured threshold in the optimal policy. This insight enables the construction of an index function that guides the sampling decision based on the current belief state.

Notably, simulation results show that the proposed index sampling policy achieves near-optimal performance, especially in moderate-to-high average delay regimes, while significantly reducing computational overhead.

II. RELATED WORK

The most relevant works on UoI are [22] and [23]. Specifically, Chen et al. [22] proposed a Whittle index-based policy for binary Markov sources, while [23] extended it to general finite-state sources over unreliable channels. Both studies considered a constant one-slot delivery delay without a sampling frequency constraint, where a binary decision—whether to sample and transmit—is made at each time step. Beyond these receiver-scheduled settings, recent studies have also investigated UoI-related metrics in random access systems, such as [25] and [29]. These works focus on slotted ALOHA channels, where sensors transmit updates probabilistically in a grant-free, push-based manner without explicit scheduling from the receiver. In contrast, our work considers a fundamentally different monitoring paradigm. We study a receiver-controlled pull-based remote monitoring system, in which the receiver explicitly determines the sampling timing under a strict long-term sampling frequency constraint. The resulting problem is formulated as a C-POSMDP, enabling more precise and deterministic management of estimation uncertainty in the presence of random transmission delays.

In the broader context of age-related optimization, several studies have examined C-SMDP and SMDP formulations under random delay in single-source systems [30], [31], [32], [33], [34], [35], [36], [37]. For instance, [32] formulated an age penalty minimization problem in continuous time, and later extensions, such as [31], addressed the discrete-time case. Ornee and Sun [34] minimized the mean square estimation error (MSEE) when observing an Ornstein–Uhlenbeck (OU) process under random delay, framed as a CMDP in continuous time with an uncountable state space. Li et al. [30] solved an age-aware remote MDP problem where actions of the source and the sampler are both decided and optimized. These works typically focus on age-based metrics that are monotonic with respect to information freshness and do not depend on the history observations of source states. An exception in this line of research is the optimization of the AoII, as studied in [37]. While AoII effectively penalizes the duration of erroneous estimation, it is a post-hoc metric that requires knowledge of the true state or focuses on error magnitude. In contrast, UoI quantifies the receiver’s real-time ambiguity about the source state based solely on history, making it intrinsically suitable for scenarios where minimizing the uncertainty of estimation is prioritized over simply counting error duration.

UoI differs from traditional freshness metrics in two key ways: it is nonmonotonic with respect to AoI, and its evolution is explicitly dependent on the content of the most recently received packet. These properties invalidate conventional index or threshold-based policies and introduce significant analytical challenges. Some recent studies [14], [15], [16], [38] explored nonmonotonic AoI penalties. They utilized SMDP or restless multiarmed bandit (RMAB) frameworks for optimal sampling design. In those works, optimal index policies remain

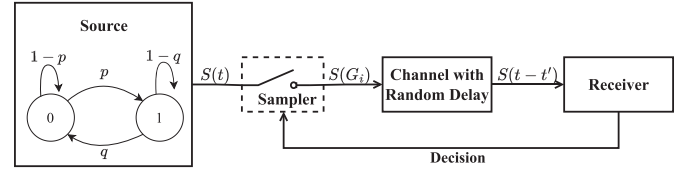


Fig. 3. System model of the considered remote estimation system.

tractable because the objective functions are independent of the actual content of the received packet. In contrast, UoI explicitly depends on the content of the received packet, making it intractable to derive exact index policies using standard approaches, which we also addressed in this article.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a discrete-time remote estimation system, as illustrated in Fig. 3, where a binary Markov source¹ is observed and controlled by a receiver. The system follows a common monitoring architecture,² where the receiver decides whether to sample the source based on past observations and sends control commands to a sampler. This centralized scheduling is well-suited for mission-critical IoT links where collision-free updates and precise uncertainty management are preferred over contention-based access. Although less scalable to massive numbers of sensors compared to grant-free protocols, it provides deterministic control over the estimation quality.

The system operates in time slots, where each slot aligns with one Markov state transition of the source. The state can only be observed when sampled. The source evolves as a discrete-time binary Markov process with state $S(t) \in \{0, 1\}$ at time t and the transition matrix

$$\begin{array}{cc} & \begin{array}{cc} \text{State 0} & \text{State 1} \end{array} \\ \begin{array}{c} \text{State 0} \\ \text{State 1} \end{array} & \begin{bmatrix} 1-p & p \\ q & 1-q \end{bmatrix} \end{array}$$

where $0 < p \leq q < 1$ and $p + q \neq 1$.³ The n -step transition matrix is [40, Appendix A]

$$\mathbf{P}^n = \begin{bmatrix} 1 - p^{(n)} & p^{(n)} \\ q^{(n)} & 1 - q^{(n)} \end{bmatrix} \quad (1)$$

with $p^{(n)} \triangleq (p - p(1 - p - q)^n)/(p + q)$, $q^{(n)} \triangleq (q - q(1 - p - q)^n)/(p + q)$, and $n \in \mathbb{N}^+$. Here, $p^{(n)}$ denotes the probability of transitioning from state 0 to 1 in n steps.

Let G_i be the time slot when the i th sample is generated and transmitted, and $R_i = G_i + Y_i$ be its reception time at the receiver, where $Y_i \in \{1, 2, \dots, Y_{\max}\}$ represents a bounded and discrete random transmission delay, with $Y_{\max}$ denoting the maximum delay in integer time slots. It is assumed to be

¹The binary source is a commonly used model (see [17], [25]), providing fundamental insights into sampling and transmission optimization.

²A general model assumes that the source evolves autonomously, and actions influence only its observability [39].

³Without loss of generality, we assume that $p \leq q$. When $p + q = 1$, UoI becomes constant and unaffected by any sampling policy [22]. Similarly, we exclude degenerate cases where $p + q = 0$ or $p + q = 2$.

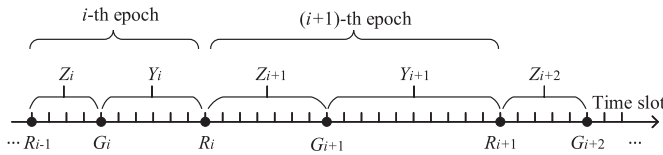


Fig. 4. Timing illustration.

an i.i.d. random variable. Physically, Y_i represents the service time, which aggregates the effects of wireless channel fading and MAC-layer retransmissions. The boundedness arises from practical protocols (e.g., truncated ARQ) that impose a maximum number of retransmissions to avoid excessive latency. The receiver sends sampling decisions to the sampler over a delay-free control link,⁴ so the decision for the i th sample is made at R_{i-1} and executed at G_i . The waiting time between sampling and last receiving is denoted as $Z_i = G_i - R_{i-1} \geq 0$. Since sampling decisions are made at R_i , this time also marks the decision epoch in our formulation. A timing diagram is shown in Fig. 4.

B. Problem Formulation

UoI, as a metric proposed in [22], quantifies the receiver's uncertainty about the source state based on observation history

$$U(t) \triangleq - \sum_{k \in \{0,1\}} \Pr[S(t) = k | \mathbf{W}(t)] \log_2 \Pr[S(t) = k | \mathbf{W}(t)] \quad (2)$$

where $\mathbf{W}(t) = (S(G_0), S(G_1), \dots, S(t'))$ denotes the observations received up to time t and $t' \triangleq \max\{G_i : R_i \leq t\}$ is the time of the most recently received update. Leveraging the Markov property of $S(t)$, we simplify

$$U(t) = - \sum_{k \in \{0,1\}} \Pr[S(t) = k | S(t')] \log_2 \Pr[S(t) = k | S(t')]. \quad (3)$$

Given the definition of t' , $U(t)$ is piecewise over each reception interval

$$U(t) = - \sum_{k \in \{0,1\}} \Pr[S(t) = k | S(G_i)] \log_2 \Pr[S(t) = k | S(G_i)] \quad \text{if } R_i \leq t < R_{i+1} \quad \forall i \in \mathbb{N}. \quad (4)$$

Define the entropy function $H(p) \triangleq -p \log_2 p - (1-p) \log_2 (1-p)$. With (1), UoI is further expressed as

$$U(t) = \begin{cases} H(p^{(t-G_i)}), & \text{if } S(G_i) = 0, R_i \leq t < R_{i+1} \\ H(q^{(t-G_i)}), & \text{if } S(G_i) = 1, R_i \leq t < R_{i+1} \end{cases} \quad (5)$$

where $t - G_i$ (for $R_i \leq t < R_{i+1}$) is the age of the latest observation. Thus, UoI is a nonmonotonic function of AoI, customized by the observed state $S(G_i)$ —unlike metrics such as nonlinear AoI or content-aware AoI.

A sampling policy $\pi = \{G_1, G_2, \dots\} \in \Pi$ is a sequence of sampling time slots. Following [32], we assume that no sample is taken, while the channel is busy, i.e., $G_{i+1} \geq R_i$, or equivalently, $G_{i+1} - Z_{i+1} = R_i$ with $Z_{i+1} \geq 0$.

⁴Our model can be extended to incorporate two-way delays.

Our objective is to minimize the time-average expected UoI under a long-term average sampling frequency constraint

$$\mathcal{P1} : \inf_{\pi \in \Pi} \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} U(t) \right] \quad (6a)$$

$$\text{s.t.} \quad \liminf_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \left[\sum_{i=0}^n (G_{i+1} - G_i) \right] \geq \frac{1}{f_{\max}} \quad (6b)$$

where $f_{\max}$ denotes the maximum allowable sampling frequency. When $f_{\max} = \infty$, the problem is unconstrained.

Problem $\mathcal{P1}$ differs fundamentally from prior works that optimize AoI-based objectives under random delay. Unlike traditional formulations, UoI depends not only on the AoI ($t - G_i$) but also on the value of the most recent observation $S(G_i)$, as shown in (5).

C. Belief-State C-POSMDP

In problem $\mathcal{P1}$, the receiver selects sampling actions based on delayed observations $\mathbf{W}(t)$, aiming to minimize the time-average UoI. This can be modeled as a C-POSMDP, which is typically addressed by converting it into a belief C-SMDP via the concept of a belief state [41]. We next define the belief state in our context and reformulate $\mathcal{P1}$ accordingly.

The belief state is defined as the conditional probability that $S(t) = 1$ given $\mathbf{W}(t)$

$$\Omega(t) = \Pr[S(t) = 1 | \mathbf{W}(t)]. \quad (7)$$

Using the Markov property and a piecewise update rule like (3) and (4), it can be shown that for $R_i \leq t < R_{i+1}$, $\forall i \in \mathbb{N}$

$$\Omega(t) = \begin{cases} p^{(t-G_i)}, & \text{if } S(G_i) = 0, R_i \leq t < R_{i+1} \\ 1 - q^{(t-G_i)}, & \text{if } S(G_i) = 1, R_i \leq t < R_{i+1} \end{cases} \quad \forall i \in \mathbb{N}. \quad (8)$$

By noting $H(p) = H(1-p)$, and combining (5) with (8), we obtain

$$U(t) = H(\Omega(t)). \quad (9)$$

Hence, $\mathcal{P1}$ can be reformulated as the following belief-based optimization problem $\mathcal{P2}$:

$$\mathcal{P2} : \bar{p}_{\text{opt}} = \inf_{\pi \in \Pi} \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} H(\Omega(t)) \right] \quad (10a)$$

$$\text{s.t.} \quad \liminf_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \left[\sum_{i=0}^n (G_{i+1} - G_i) \right] \geq \frac{1}{f_{\max}} \quad (10b)$$

where $\bar{p}_{\text{opt}}$ denotes the optimal value of $\mathcal{P2}$.

Problem $\mathcal{P2}$, thus, becomes a C-SMDP over an infinite horizon [42], [43]. Due to the **state dependence** property of UoI, solving the Bellman equations becomes nontrivial, motivating a reformulation approach.

D. Properties of Belief State

Before reformulating the problem, we introduce the evolution and key properties of the belief state $\Omega(t)$, inspired by results in [22]. These properties provide useful insights for analyzing state transitions and designing sampling policies. The belief state evolves as follows.

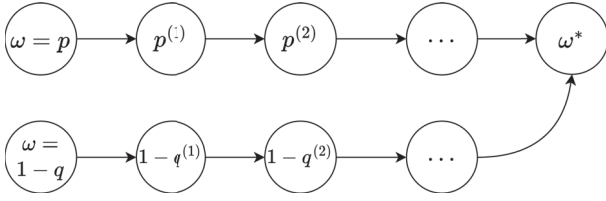


Fig. 5. Two evolution cases of belief state without observation updates based on Lemma 1 and Corollary 3.

Lemma 1: For $R_i \leq t < R_{i+1}$, $\forall i \in \mathbb{N}$, given $\Omega(t) = \omega$, the belief state at time $t + k$ is given by

$$\Omega(t + k) = \frac{p - p(1 - p - q)^k}{p + q} + \omega(1 - p - q)^k \quad (11)$$

where $\omega \in \{p^{(n)}, 1 - q^{(n)}\}$, $n \in \mathbb{N}$, and $0 \leq k < R_{i+1} - t$. For brevity, we denote

$$\tau^k(\omega) \triangleq \Omega(t + k). \quad (12)$$

Proof: See [1, Appendix A]. $\square$

The belief state evolution satisfies the following properties.

Corollary 1: For any $n \in \mathbb{N}$: If $\omega = p^{(n)}$, then $\tau^k(\omega) = p^{(n+k)}$. If $\omega = 1 - q^{(n)}$, then $\tau^k(\omega) = 1 - q^{(n+k)}$.

Proof: This result follows directly from (11). $\square$

Corollary 2: For any $n \in \mathbb{N}$, it holds that $0 < p^{(n)} < 1$ and $0 < 1 - q^{(n)} < 1$.

Proof: See Appendix A. $\square$

Corollary 3: The equilibrium belief state of $\Omega(t)$ is given by

$$\omega^* \triangleq \lim_{k \rightarrow \infty} \tau^k(\omega) = \frac{p}{p + q} \quad (13)$$

which holds when $R_{i+1} - t = \infty$, i.e., no observations are received after time t .

Proof: Since $0 < |1 - p - q| < 1$, we have $\lim_{k \rightarrow \infty} (1 - p - q)^k = 0$, yielding the desired result. $\square$

Corollary 1 describes the deterministic evolution of the belief state, allowing direct computation of the next state. Corollary 2 guarantees belief values remain in $(0, 1)$, supporting convergence proof of the RVI algorithm in Theorem 2. Corollary 3 confirms the existence of a steady-state belief, which motivates the approximation in Section V-A.

A graphical illustration of belief state evolution is provided in Fig. 5. We can observe that any initial belief $\omega \in \{p^{(n)}, 1 - q^{(n)}\}$ asymptotically converges to the equilibrium value ω^* in the absence of future observation updates.

IV. OPTIMAL SAMPLING POLICY

This section derives the optimal sampling policy for minimizing the UoI under a sampling frequency constraint. This optimal design, detailed in this section, paves the way for the suboptimal index-based sampling policy in Section V.

A. Reformulation of $\mathcal{P}2$

As discussed earlier, to facilitate tractable analysis, we reformulate the C-SMDP into a simpler equivalent problem.

Proposition 1: A sampling policy $\pi = (G_1, G_2, \dots) \in \Pi$ is equivalently represented as $\pi = (Z_1, Z_2, \dots) \in \Pi$.

Proof: The sampling policy can be expressed via waiting times $\pi = (Z_1, Z_2, \dots)$. Let $G_0 = 0$, and define $G_i = \sum_{j=0}^{i-1} (Y_j + Z_{j+1})$ and $R_i = G_i + Y_i$ for each $i \geq 0$. Given the sequence $(Y_0, Y_1, \dots)$, the sampling instants $(G_1, G_2, \dots)$ are uniquely determined by $(Z_1, Z_2, \dots)$. Hence, representing π in terms of Z_i is valid. $\square$

Proposition 2: Given $\tau^k(\Omega(R_i)) = \Omega(R_i + k)$ and $\Omega(R_i) = \omega$, $\mathcal{P}2$ can be reformulated as

$$\mathcal{P}3: \bar{p}_{\text{opt}} = \inf_{\pi \in \Pi} \lim_{n \rightarrow \infty} \frac{\sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1} + Y_{i+1} - 1} H(\tau^k(\omega)) \right]}{\sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1} + Y_{i+1}]} \quad (14a)$$

$$\text{s.t. } \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1} + Y_{i+1}] \geq \frac{1}{f_{\max}} \quad (14b)$$

Proof: With $\tau^k(\Omega(R_i)) = \Omega(R_i + k)$ and $\Omega(R_i) = \omega$, we obtain

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} H(\Omega(t)) \right] \quad (15a)$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{t=R_i}^{R_{i+1}-1} H(\Omega(t)) \right]}{\sum_{i=0}^{n-1} \mathbb{E} [R_{i+1} - R_i]} \quad (15b)$$

$$= \lim_{n \rightarrow \infty} \frac{\sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1} + Y_{i+1} - 1} H(\tau^k(\omega)) \right]}{\sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1} + Y_{i+1}]} \quad (15c)$$

To transform the constraint in (10b), note that $Z_{i+1} = G_{i+1} - R_i = G_{i+1} - (G_i + Y_i)$, implying that $Z_{i+1} + Y_i = G_{i+1} - G_i$. Since $\{Y_i\}$ are i.i.d., we have $\mathbb{E}[Z_{i+1} + Y_{i+1}] = \mathbb{E}[G_{i+1} - G_i]$. $\square$

Problem $\mathcal{P}3$ is a nonlinear fractional program, which is challenging to solve due to its ratio form. To address this, we adopt a sequential decision process incorporating the Dinkelbach parameter $\beta \geq 0$

$\mathcal{P}4:$

$$S(\beta) \triangleq \inf_{\pi \in \Pi} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \left\{ \mathbb{E} \left[\sum_{k=0}^{Z_{i+1} + Y_{i+1} - 1} H(\tau^k(\omega)) \right] - \beta \mathbb{E}[Z_{i+1} + Y_{i+1}] \right\} \quad (16a)$$

$$= \inf_{\pi \in \Pi} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1} + Y_{i+1} - 1} [H(\tau^k(\omega)) - \beta] \right] \quad (16b)$$

$$\text{s.t. } \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1} + Y_{i+1}] \geq \frac{1}{f_{\max}}. \quad (16c)$$

Applying Dinkelbach's method for nonlinear fractional programming [28], we obtain the following lemma.

Lemma 2: There are three assertions that hold true as follows.

- 1) $\bar{p}_{\text{opt}} \leq \beta$ if and only if $S(\beta) \leq 0$.
- 2) $\bar{p}_{\text{opt}}$ is the unique root of $S(\beta) = 0$.
- 3) If $S(\beta) = 0$, the solutions to $\mathcal{P}3$ and $\mathcal{P}4$ are identical.

Proof: See Appendix B. $\square$

This lemma implies that solving $\mathcal{P}3$ reduces to finding the root of $S(\beta)$ such that $S(\bar{p}_{\text{opt}}) = 0$, thereby identifying the optimal value $\bar{p}_{\text{opt}}$ through $\mathcal{P}4$.

B. CMDP Reformulation and Lagrangian Dual Approach

Through four transformation steps, solving $\mathcal{P}1$ reduces to solving $\mathcal{P}4$. Given $\bar{p}_{\text{opt}}$, this becomes a CMDP, whose key components are analyzed as follows.

- 1) *State Space*: The belief state $\Omega(R_i)$ at sampling time R_i belongs to $\mathcal{S} = \{1 - q^{(y)}, p^{(y)} : y \in \mathcal{Y}\}$ due to the i.i.d. delay Y_i . It captures the receiver's estimate of the Markov source based on delayed observations Y_i time steps ago.
- 2) *Action Space*: The waiting time Z_i for all $i \in \mathbb{N}^+$, $Z_i \in \mathcal{Z} = \{0, 1, 2, \dots, M\}$.
- 3) *Transition Probability*: The next state $\Omega(R_{i+1})$ depends on state $\Omega(R_i)$, waiting time Z_{i+1} , and delay Y_{i+1}

$$\begin{aligned} P[\Omega(R_{i+1}) | \Omega(R_i)] \\ = \begin{cases} \tau^{Z_{i+1}+Y_{i+1}}(\Omega(R_i)), & \text{if } \Omega(R_{i+1}) = 1 - q^{(Y_{i+1})} \\ 1 - \tau^{Z_{i+1}+Y_{i+1}}(\Omega(R_i)), & \text{if } \Omega(R_{i+1}) = p^{(Y_{i+1})}. \end{cases} \end{aligned} \quad (17)$$

- 4) *Cost Function*: Based on the objective in $\mathcal{P}4$, the per-stage cost is

$$\begin{aligned} C_i(Z_{i+1}, \Omega(R_i); \bar{p}_{\text{opt}}) \\ = \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\Omega(R_i+k)) - \bar{p}_{\text{opt}}] \right]. \end{aligned} \quad (18)$$

To solve $\mathcal{P}4$, we apply the Lagrangian dual approach, targeting the computation of $S(\bar{p}_{\text{opt}})$. We convert the CMDP into an unconstrained MDP by introducing the Lagrangian dual variable $\lambda \geq 0$.

Define the Lagrangian average cost criterion as

$$\Phi(\pi; \alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{E}^\pi \left[\sum_{i=0}^{n-1} C_i(Z_{i+1}, \Omega(R_i); \alpha) \right] \quad (19)$$

where $C_i(Z_{i+1}, \Omega(R_i); \alpha)$ is the Lagrangian per-interval cost

$$\begin{aligned} C_i(Z_{i+1}, \Omega(R_i); \alpha) \\ = \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\Omega(R_i+k)) - (\bar{p}_{\text{opt}} + \lambda)] \right] \\ = \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\Omega(R_i+k)) - \alpha] \right] \end{aligned} \quad (20)$$

for $\alpha = \bar{p}_{\text{opt}} + \lambda$, and the dual value $\lambda \geq 0$.

The resulting infinite-horizon MDP is called *Lagrangian MDP* [41, Section 6.6.2], denoted by

$$\mathcal{P}5: \quad \min_{\pi \in \Pi} \Phi(\pi; \alpha). \quad (21)$$

Let π_α^* denote an optimal deterministic policy for (21). The following theorem characterizes the optimality conditions.

Theorem 1: Given α and $\Omega(R_i) = \omega$, the optimal policy $\pi_\alpha^* = (Z_1^*(\alpha), Z_2^*(\alpha), \dots)$ to $\mathcal{P}5$ satisfies Bellman's equation

$$s(\alpha) + \tilde{V}(\omega, \alpha) = \inf_{Z_{i+1} \in \mathcal{Z}} \{C_i(Z_{i+1}, \omega; \alpha) + r(Z_{i+1}, \omega; \alpha)\} \quad (22)$$

$$s(\alpha) = \inf_{Z_{i+1} \in \mathcal{Z}} \{C_i(Z_{i+1}, p; \alpha) + r(Z_{i+1}, p; \alpha)\} \quad (23)$$

where

$$\begin{aligned} r(Z_{i+1}, \omega; \alpha) = \mathbb{E} \left[\tau^{Z_{i+1}+Y_{i+1}}(\omega) \tilde{V}(1 - q^{(Y_{i+1})}, \alpha) \right. \\ \left. + (1 - \tau^{Z_{i+1}+Y_{i+1}}(\omega)) \tilde{V}(p^{(Y_{i+1})}, \alpha) \right]. \end{aligned} \quad (24)$$

Proof: See Appendix C. $\square$

Having solved the Lagrangian MDP $\mathcal{P}5$, we now examine the relationship between the primal and dual problems. Define

$$\phi(\alpha) \triangleq \Phi(\pi_\alpha^*; \alpha) + \frac{\alpha - \bar{p}_{\text{opt}}}{f_{\text{max}}}. \quad (25)$$

The Lagrangian dual problem of $\mathcal{P}4$ is denoted as

$$d \triangleq \sup_{\alpha \geq 0} \phi(\alpha) \quad (26)$$

where d is the optimal dual value of (26).

By weak duality [44, Prop. 5.2.2], we have $d \leq S(\bar{p}_{\text{opt}})$. To establish strong duality ($d = S(\bar{p}_{\text{opt}})$), we invoke the Karush–Kuhn–Tucker (KKT) conditions [44, Chapter 5.5.3].

Lemma 3: The duality gap is zero, and the pair (π_α^*, α^*) is optimal if and only if the KKT conditions are satisfied

$$\alpha^* \geq \bar{p}_{\text{opt}} \quad (27a)$$

$$\pi_{\alpha^*}^* = \arg \min_{\pi} \Phi(\pi; \alpha^*) \quad (27b)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}^{\pi_{\alpha^*}^*} [Z_{i+1} + Y_{i+1}] - \frac{1}{f_{\text{max}}} \geq 0 \quad (27c)$$

$$(\alpha^* - \bar{p}_{\text{opt}}) \left\{ \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}^{\pi_{\alpha^*}^*} [Z_{i+1} + Y_{i+1}] - \frac{1}{f_{\text{max}}} \right\} = 0. \quad (27d)$$

Discussion 1: If such a pair $(\pi_{\alpha^*}^*, \alpha^*)$ exists, it achieves optimality and zero duality gap, i.e., $d = S(\bar{p}_{\text{opt}})$. The remaining task is to find such a pair satisfying (27a)–(27d). See Appendix D for a detailed explanation.

C. Three-Step Solutions of Optimal Policies

1) Step One: Sampling Without Frequency Constraints:

Assume that $\alpha = \bar{p}_{\text{opt}}$; then, we have $s(\alpha) = \inf_{\pi \in \Pi} \Phi(\pi; \alpha) = S(\bar{p}_{\text{opt}}) = 0$. The optimal solution can, thus, be obtained by finding the root of $s(\alpha)$, as described in **Step 1** of Algorithm 1.

2) *Step Two: Feasibility Check of the Policy*: For simplification, denote the frequency constraint as

$$h(\pi_\alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}^{\pi_\alpha} [Z_{i+1} + Y_{i+1}] - \frac{1}{f_{\text{max}}}. \quad (28)$$

If α and π_α satisfy frequency constraints, i.e., $h(\pi_\alpha) \geq 0$, then $(\pi_{\alpha^*}^*, \alpha^*) = (\pi_{\bar{p}_{\text{opt}}}^*, \bar{p}_{\text{opt}})$. Otherwise, go to *Step Three*.

Algorithm 1 Bisec-RVI Algorithm

Input: tolerance $\epsilon > 0$
Output: the deterministic policy (policies) for $\mathcal{P}5$

Step 1. Sampling without frequency constraints;

- 1 Initialize: $l = 0, u = 1$;
- 2 **while** $u - l \geq \epsilon$ **do**
- 3 $\alpha := (l + u)/2$;
- 4 Run RVI to solve $\mathcal{P}5$, compute $s(\alpha)$ and $\tilde{V}(\omega, \alpha)$;
- 5 **if** $s(\alpha) > 0$ **then**
- 6 $l := \alpha$;
- 7 **else**
- 8 $u := \alpha$;

9 Proceed to Step 2 with α and π_α ;

Step 2. Check feasibility of the policy from Step 1;

- 10 **if** $h(\pi_\alpha) \geq 0$ **then**
- 11 **return** $\bar{p}_{\text{opt}} = \alpha^* = \alpha$ and $\pi_{\alpha^*} = \pi_\alpha$
- 12 **else**
- 13 Proceed to Step 3;

Step 3. Sampling with frequency constraints;

- 14 Initialize $\alpha_1 = 0, \alpha_2 = 1$;
- 15 **while** $\alpha_2 - \alpha_1 \geq \epsilon$ **do**
- 16 $\alpha := (\alpha_1 + \alpha_2)/2$;
- 17 Run RVI to compute $s(\alpha)$ and $\tilde{V}(\omega, \alpha)$;
- 18 **if** $h(\pi_\alpha) > 0$ **then**
- 19 $\alpha_2 := \alpha$;
- 20 **else if** $h(\pi_\alpha) < 0$ **then**
- 21 $\alpha_1 := \alpha$;
- 22 **else**
- 23 $\alpha_2 = \alpha_1 := \alpha$;

24 **return** $\alpha^* = \alpha, \pi_{\alpha_1}^*$ and $\pi_{\alpha_2}^*$.

3) *Step Three: Sampling Under Frequency Constraints:* If $\alpha = \bar{p}_{\text{opt}}$ violates the frequency constraint, it indicates that the constraint is active, i.e., $\alpha^* > \bar{p}_{\text{opt}}$ and $h(\pi_{\alpha^*}) = 0$. We first characterize the behavior of $h(\pi_\alpha)$.

Lemma 4: The function $h(\pi_\alpha)$ has the following properties.

- 1) It is nondecreasing in α .
- 2) $\lim_{\alpha \rightarrow 0} h(\pi_\alpha) \leq 0$ and $h(\pi_1) > 0$, provided that $h(\pi_{\alpha^*}) = 0$.

Proof: See Appendix E. $\square$

Using these properties, we find the root of $h(\pi_\alpha)$ by binary search as in **Step 3** of Algorithm 1. To construct the final policy, we compute the randomized mixture component c between two deterministic optimal policies $\pi_{\alpha_1}^*$ and $\pi_{\alpha_2}^*$. Let

$$\Delta_{\alpha_1} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}^{\pi_{\alpha_1}^*} [Z_{i+1} + Y_{i+1}] \quad (29)$$

$$\Delta_{\alpha_2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}^{\pi_{\alpha_2}^*} [Z_{i+1} + Y_{i+1}]. \quad (30)$$

If $\Delta_{\alpha_1} \neq \Delta_{\alpha_2}$, the mixture component is given by

$$c = \frac{1/f_{\max} - \Delta_{\alpha_2}}{\Delta_{\alpha_1} - \Delta_{\alpha_2}}. \quad (31)$$

If $\Delta_{\alpha_1} = \Delta_{\alpha_2}$, then $c = 1$; the optimal policy is deterministic.

Lemma 5: (Restatement of Theorem 6.6.2 in [41]) When constraint is active, the optimal policy for $\mathcal{P}4$ is a randomized mixture of two deterministic policies of the Lagrangian MDP

$$\pi_{\alpha^*}^* = c\pi_{\alpha_1}^* + (1 - c)\pi_{\alpha_2}^*, \quad \text{where } c \in [0, 1].$$

Lemma 6: The value of $\bar{p}_{\text{opt}}$ is given as follows:

- 1) $\bar{p}_{\text{opt}} = \alpha^*$ if the constraint is inactive;
- 2) $\bar{p}_{\text{opt}} = f_{\max} s(\alpha^*) + \alpha^*$ if the constraint is active.

Proof: See Appendix F. $\square$

Up until now, one key issue remains unresolved: since the RVI algorithm is applied in Algorithm 1 to solve problem $\mathcal{P}5$, it is essential to establish the convergence of this iterative procedure. This ensures that the derived solution is not only theoretically grounded but also practically computable.

Theorem 2: [Convergence of RVI] Given $0 < p \leq q < 1$, $0 < p + q < 2$ and $p + q \neq 1$, the RVI algorithm is convergent.

Proof: See Appendix G. $\square$

The conditions in Theorem 2 guarantee that UoI is time-varying but not constant. This result ensures the reliability of Algorithm 1 in obtaining a stable and optimal policy. The proof relies on the property of belief state in Corollary 2.

V. SUBOPTIMAL INDEX-BASED POLICY

While optimal, the policy for $\mathcal{P}1$ demands repeated RVI, making it computationally expensive. This section introduces a low-complexity, index-based threshold policy that provides near-optimal performance without iterative value updates.

A. Approximation of Bellman

To design a computation-efficient policy, we try to use an approximation for Bellman's equations. According to the equilibrium property of belief state in Corollary 3, we have

$$\begin{aligned} \lim_{Y_{i+1} \rightarrow \infty} \Pr \left[\Omega(R_{i+1}) = 1 - q^{(Y_{i+1})} | \Omega(R_i) \right] \\ = \tau^{Z_{i+1} + Y_{i+1}}(\Omega(R_i)) = \omega^*. \end{aligned} \quad (32)$$

Therefore, if the transmission delay Y_i is sufficiently large^{5,6} the transition probability $\Pr[\Omega(R_{i+1}) = 1 - q^{(Y_{i+1})}]$ can be well approximated by a constant value ω^* . This approximation simplifies the Bellman equation, yielding a tractable analytical structure that enables the design of an efficient index-based sampling policy.

Based on the large delay assumption, we formulate the following optimization problem:

$$\mathcal{P}6: \quad \bar{p}_{\text{nopt}} = \inf_{\psi \in \Pi} \limsup_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \left[\sum_{t=0}^{T-1} H(\Omega(t)) \right] \quad (33a)$$

$$\text{s.t. } \liminf_{n \rightarrow \infty} \frac{1}{n} \mathbb{E} \left[\sum_{i=1}^n (G_{i+1} - G_i) \right] \geq \frac{1}{f_{\max}}. \quad (33b)$$

We use the optimal solution to $\mathcal{P}6$ to construct a suboptimal index-based policy for $\mathcal{P}1$. The reformulation of $\mathcal{P}6$ follows

⁵This assumption is applicable to long-distance communication scenarios such as nonterrestrial networks.

⁶This assumption is applicable to DTNs, which often arise in long-distance communication scenarios.

the same structure as in Section IV-A. Skipping analogous steps to Propositions 1 and 2, we directly adopt a sequential decision formulation with the Dinkelbach parameter $\beta \geq 0$

$\mathcal{P}7$:

$$H(\beta) \triangleq \inf_{\psi \in \Pi} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1} + Y_{i+1} - 1} [H(\tau^k(\omega)) - \beta] \right] \quad (34a)$$

$$\text{s.t. } \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[Z_{i+1} + Y_{i+1}] \geq \frac{1}{f_{\max}}. \quad (34b)$$

We then establish the equivalence between $\mathcal{P}6$ and $\mathcal{P}7$ by the following lemma.

Lemma 7: The following statements hold.

- 1) $\bar{\rho}_{\text{nopt}} \leq \beta$ if and only if $H(\beta) \leq 0$.
- 2) $\bar{\rho}_{\text{nopt}}$ is the unique root of $H(\beta) = 0$.
- 3) If $H(\beta) = 0$, the solutions to $\mathcal{P}6$ and $\mathcal{P}7$ are identical.

Proof: The proof follows directly from Lemma 2. $\square$

This lemma allows us to solve $\mathcal{P}6$ by finding the unique root of $H(\beta)$ such that $\beta = \bar{\rho}_{\text{nopt}} \geq 0$ and $H(\bar{\rho}_{\text{nopt}}) = 0$.

B. CMDP Reformulation and Lagrangian Dual Approach

Following the reformulation, the optimal policy for $\mathcal{P}6$ can be determined by solving $\mathcal{P}7$. Therefore, our current goal is to solve $\mathcal{P}7$ given $\bar{\rho}_{\text{nopt}}$, which corresponds to a CMDP. The core elements of this CMDP are outlined in the following.

- 1) *State Space:* The belief state $\Omega(R_i)$ at time R_i lies $\Omega(R_i) \in \mathcal{S} = \{1 - q^{(y)}, p^{(y)} : y \in \mathcal{Y}\}$ for all $i \in \mathbb{N}$.
- 2) *Action Space:* The waiting time Z_i with $Z_i \in \mathcal{Z} = \{0, 1, 2, \dots, M\}$ for all $i \in \mathbb{N}^+$.
- 3) *Transition Probability:* The state transition is

$$\begin{aligned} P[\Omega(R_{i+1}) | \Omega(R_i)] \\ = \begin{cases} \omega^*, & \text{if } \Omega(R_{i+1}) = 1 - q^{(Y_{i+1})} \\ 1 - \omega^*, & \text{if } \Omega(R_{i+1}) = p^{(Y_{i+1})}. \end{cases} \end{aligned} \quad (35)$$

- 4) *Cost Function:* The cost function is defined as follows:

$$\begin{aligned} C_i(Z_{i+1}, \Omega(R_i); \bar{\rho}_{\text{nopt}}) \\ = \mathbb{E} \left[\sum_{k=0}^{Z_{i+1} + Y_{i+1} - 1} [H(\Omega(R_i + k)) - \bar{\rho}_{\text{nopt}}] \right]. \end{aligned} \quad (36)$$

Remark 1: The main differences between $\mathcal{P}7$ and $\mathcal{P}4$ lie in the **transition probability** and **cost function**. Under the large delay assumption, state transitions become independent of $\Omega(R_i)$, simplifying the problem and enabling the design of index-based policies. However, while the resulting policy is optimal for $\mathcal{P}6$, it is generally suboptimal for $\mathcal{P}4$, i.e., $\bar{\rho}_{\text{nopt}} \neq \bar{\rho}_{\text{opt}}$.

To solve $\mathcal{P}7$, we apply the Lagrangian duality framework and reformulate the CMDP as an unconstrained MDP. We define the following Lagrangian MDP:

$$\mathcal{P}8: \quad \min_{\psi \in \Pi} \Phi(\psi; \rho) \quad (37)$$

where $\Phi(\cdot; \cdot)$ is defined in (19) and (20), and $\rho = \bar{\rho}_{\text{nopt}} + \lambda$ with $\lambda \geq 0$ being the dual variable. Let ψ_ρ^* denote the optimal policy for $\mathcal{P}8$. It can be explicitly characterized as follows.

Theorem 3: Given ρ , the optimal policy $\psi_\rho^* = (Z_1^*(\rho), Z_2^*(\rho), \dots)$ for $\mathcal{P}7$ is expressed as

$$Z_{i+1}^*(\rho) = \min \{k \in \mathcal{Z} : \eta(\Omega(t+k)) \geq \rho, t = R_i\} \quad (38)$$

where $\eta(\cdot)$ is an index function defined by

$$\eta(\omega) \triangleq \inf_{z \in \mathbb{N}^+} \frac{1}{z} \sum_{j=0}^{z-1} \mathbb{E}[H(\tau^{j+Y_i}(\omega))]. \quad (39)$$

Proof: See Appendix H. $\square$

Remark 2: Compared to Theorem 1, the optimal policy ψ_ρ^* can be obtained directly from the index function without running the RVI algorithm, greatly simplifying computation.

We now analyze the connection between the primal and dual formulations. Define the dual objective

$$\phi(\rho) \triangleq \Phi(\psi_\rho^*; \rho) + \frac{\rho - \bar{\rho}_{\text{nopt}}}{f_{\max}}. \quad (40)$$

Then, the Lagrangian dual of $\mathcal{P}7$ is given by

$$\theta \triangleq \sup_{\rho \geq 0} \phi(\rho) \quad (41)$$

where θ denotes the optimal dual value.

By the weak duality theorem [44, Prop. 5.2.2], we have $\theta \leq H(\bar{\rho}_{\text{nopt}})$. To ensure strong duality, i.e., $\theta = H(\bar{\rho}_{\text{nopt}})$, the KKT conditions must be satisfied. Following Lemma 3 and Discussion 1, we propose a three-step algorithm to find the optimal ψ_ρ^* and corresponding Lagrangian multiplier ρ^* .

C. Three-Step Solutions of Suboptimal Policies

1) Step One: Sampling Without Frequency Constraints:

Assume that $\rho = \bar{\rho}_{\text{nopt}}$, and the frequency constraint is inactive. Under this setting, we have the following.

Lemma 8: If the frequency constraint is inactive, then ρ is the unique root of $f(\rho) = 0$, where

$$\begin{aligned} f(\rho) \\ = \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}(\rho) + Y_{i+1} - 1} H(\Omega(R_i + k)) \right] - \rho \mathbb{E}[Z_{i+1}(\rho) + Y_{i+1}]. \end{aligned} \quad (42)$$

Proof: See unconstrained cases in [1, Appendix C]. $\square$

Consequently, ρ can be determined by solving $f(\rho) = 0$, corresponding to **Step 1** of Algorithm 2.

2) *Step Two: Feasibility Check of the Policy:* Using the frequency constraint function $h(\psi_\rho)$ from (28), if ψ_ρ from Step 1 satisfies $h(\psi_\rho) \geq 0$, then the optimal pair is $(\psi_\rho^*, \rho^*) = (\psi_{\bar{\rho}_{\text{nopt}}}^*, \bar{\rho}_{\text{nopt}})$. Otherwise, proceed to Step 3.

3) *Step Three: Sampling Under Frequency Constraints:* If $\rho = \bar{\rho}_{\text{nopt}}$ violates the constraint, it implies that the constraint is active and $\rho^* > \bar{\rho}_{\text{nopt}}$. In this case, the optimal ρ^* satisfies $h(\psi_{\rho^*}^*) = 0$. Based on Lemma 4, we find the root of $h(\psi_\rho)$ for $\rho > 0$ via bisection, as described in **Step 3** of Algorithm 2.

Algorithm 2 Index-Based Algorithm

Input: tolerance $\epsilon > 0$
Output: the value (values) of ρ

Step 1. Sampling without frequency constraints;

```

1 Initialize:  $l = 0, u = 1$ ;
2 while  $u - l \geq \epsilon$  do
3    $\rho := (l + u)/2$ ;
4   Compute  $\psi_\rho$  by (38) and (39);
5   if  $f(\rho) > 0$  then
6      $l := \rho$ ;
7   else
8      $u := \rho$ ;
9 Proceed to Step 2 with  $\rho$ ;

Step 2. Check feasibility of the policy from Step 1;
10 if  $h(\psi_\rho) \geq 0$  then
11   return  $\bar{p}_{\text{nopt}} = \rho^* = \rho$ .
12 else
13   Proceed to Step 3;

Step 3. Sampling under frequency constraints;
14 Initialize  $\rho_1 = 0, \rho_2 = 1$ ;
15 while  $\rho_2 - \rho_1 \geq \epsilon$  do
16    $\rho := (\rho_1 + \rho_2)/2$ ;
17   Compute  $\psi_\rho$  by (38) and (39);
18   if  $h(\psi_\rho) > 0$  then
19      $\rho_2 := \rho$ ;
20   else if  $h(\psi_\rho) < 0$  then
21      $\rho_1 := \rho$ ;
22   else
23      $h(\psi_\rho) = 0$ 
24      $\rho_2 = \rho_1 := \rho$ ;
25 return  $\rho_1^*$  and  $\rho_2^*$ .
```

To construct the final policy, we compute the randomized mixture component b . Given ρ_1 and ρ_2 , define

$$\Delta_{\rho_1} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1}(\rho_1) + Y_{i+1}] \quad (43)$$

$$\Delta_{\rho_2} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1}(\rho_2) + Y_{i+1}]. \quad (44)$$

If $\Delta_{\rho_1} \neq \Delta_{\rho_2}$, then

$$b = \frac{1/f_{\max} - \Delta_{\rho_2}}{\Delta_{\rho_1} - \Delta_{\rho_2}}. \quad (45)$$

Otherwise, the optimal policy is deterministic, and set $b = 1$.

Lemma 9: If the frequency constraint is active, the optimal policy for $\mathcal{P}7$ is a randomized mixture of two deterministic policies of the Lagrangian MDP $\mathcal{P}8$: $\psi_{\rho^*} = b\psi_{\rho_1^*} + (1-b)\psi_{\rho_2^*}$, where $b \in [0, 1]$.

We compare the computational complexity of the two proposed algorithms. Recall that $|S|$ and $|Z|$ denote the cardinality of the state space and action space, respectively. Algorithm 1 requires solving an unconstrained MDP in each

TABLE I
COMPARISON OF THE PROPOSED POLICIES

Algorithm	Optimality	Time complexity
Bisec-RVI	Optimal	$\mathcal{O}(K_{bi} \cdot K_{rvi} \cdot S ^2 Z)$
Index-based	Sub-optimal	$\mathcal{O}(S \cdot Z)$

bisection step. Assuming $K_{bi} \approx \log_2(1/\epsilon)$ iterations for the bisection search and K_{rvi} iterations for RVI convergence, the total computational complexity is $\mathcal{O}(K_{bi} \cdot K_{rvi} \cdot |S|^2 |Z|)$. This iterative procedure can be computationally prohibitive for resource-constrained devices. In contrast, Algorithm 2 avoids the iterative value updates of RVI. The index function $\eta(\omega)$ in (39) is derived through explicit calculation. The complexity of constructing the index table is approximately $\mathcal{O}(|S| \cdot |Z|)$. Although a bisection search is still used to find ρ , the dominant iterative cost K_{rvi} is eliminated, making the index-based policy more efficient for practical implementation. A comparison of the proposed policies is provided in Table I.

VI. NUMERICAL RESULTS

This section presents numerical results to evaluate the performance of both the optimal and suboptimal index-based threshold policies. Following the observation in [22, Fig. 3] that belief state evolution differs significantly for $p + q < 1$ and $p + q > 1$, we consider both cases in our experiments.

We compare the following five sampling policies.

1. *UoI-Optimal*: The optimal policy in Section IV-C.
2. *Index-Based*: The suboptimal policy in Section V-C.
3. *Zero-Wait*: Sampling immediately after each packet is received ($Z_i = 0$ for $\forall i \in \mathbb{N}$), feasible only when $f_{\max} \geq 1/\mathbb{E}[Y_i]$. This policy minimizes delay and maximizes throughput.
4. *AoI-Optimal*: The AoI-minimizing policy derived from [31, Theorem 4], designed for minimizing age in discrete-time systems.
5. *Uniform*: A randomized periodic policy satisfying $\lim_{n \rightarrow \infty} \sum_{i=0}^n (G_{i+1} - G_i) = 1/f_{\max}$. Since $1/f_{\max}$ may not be an integer, the policy randomly switches between two deterministic waiting times

$$Z_{\text{uni},1} = \lceil 1/f_{\max} \rceil, Z_{\text{uni},2} = \lfloor 1/f_{\max} \rfloor$$

with mixing probability

$$u = \frac{1/f_{\max} - Z_{\text{uni},1}}{Z_{\text{uni},2} - Z_{\text{uni},1}}.$$

Fig. 6 compares five policies under $p + q < 1$. In both left and right figures, the index-based policy closely matches the performance of the UoI-optimal policy. In contrast, the AoI-optimal and uniform policies only perform competitively when $f_{\max}$ is sufficiently large. This indicates that the suboptimal index-based policy remains effective across the range of delay parameters tested when $p + q < 1$.

Fig. 7 illustrates performance under $p + q > 1$. In the left figure, the index-based policy performs comparably to the optimal one when $f_{\max} \leq 0.6$ but degrades as $f_{\max}$ increases. This degradation stems from the threshold-based nature of the

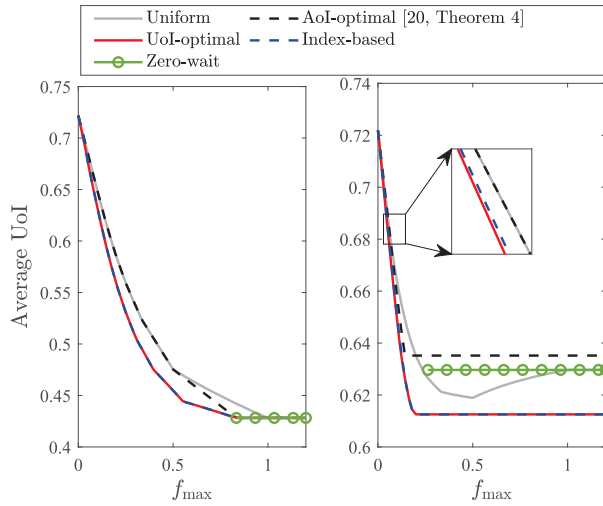


Fig. 6. Average UoI versus $f_{\max}$ with i.i.d. random delay. Left: $P[Y_i = 1] = 0.8$ and $P[Y_i = 2] = 0.2$. Right: $P[Y_i = 1] = 0.8$ and $P[Y_i = 15] = 0.2$. Source transition parameters: $p = 0.05$ and $q = 0.2$.

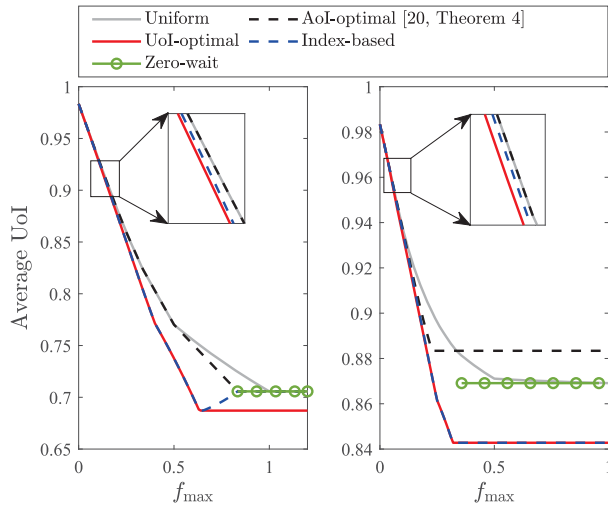


Fig. 7. Average UoI versus $f_{\max}$ under i.i.d. random delays. Left: $P[Y_i = 1] = 0.8$ and $P[Y_i = 2] = 0.2$. Right: $P[Y_i = 1] = 0.8$ and $P[Y_i = 10] = 0.2$. Source transition parameters: $p = 0.7$ and $q = 0.95$.

index policy: as $f_{\max}$ grows, the index value remains below the threshold $\bar{p}_{\text{nopt}}$, causing the policy to trigger sampling at every decision epoch and effectively degenerate into a zero-wait strategy when $f_{\max} \geq 0.83$.

The performance deterioration observed in the index-based policy is driven by the interplay between the source dynamics and the policy's approximation design. When $p + q > 1$, the UoI curve exhibits nonmonotonic valleys with respect to AoI. An optimal strategy should exploit these valleys by maintaining a specific nonzero waiting time. However, the index-based policy is derived under a large-delay approximation, which tends to underestimate the benefit of waiting in this specific regime. As $f_{\max}$ increases from 0.6 to 0.83, the relaxation of the frequency constraint allows the policy to sample more frequently. The policy utilizes this budget to reduce the waiting time Z aggressively. Specifically, the optimal policy collapses to a deterministic waiting strategy with a fixed waiting time

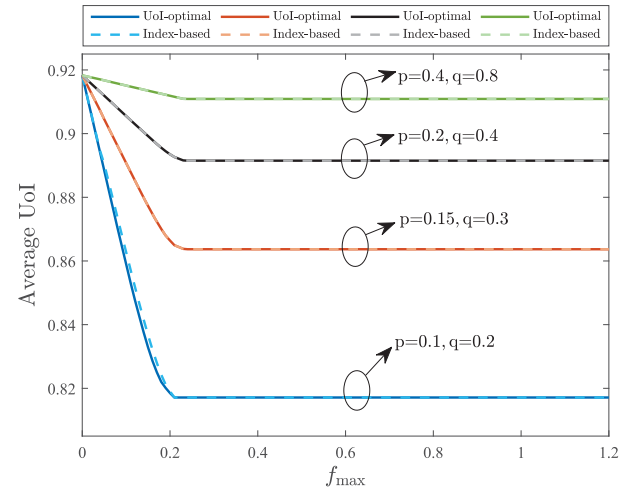


Fig. 8. Average UoI versus $f_{\max}$ with i.i.d. delay with $P[Y_i = 1] = 0.8$ and $P[Y_i = 15] = 0.2$, for varying values of p and q .

$Z = 1$, while the index-based policy implements a randomized strategy that mixes between $Z = 1$ and $Z = 0$. This leads to premature sampling—updating before the belief state can evolve into the low-uncertainty valley. Consequently, the performance deteriorates as the policy transitions from a forced-wait state toward the zero-wait strategy, failing to retain the optimal waiting intervals.

In the right figure of Fig. 7, where delays are larger, the index-based policy again tracks the optimal one closely, confirming its robustness under high-latency conditions.

Fig. 8 further evaluates the index-based policy under various source dynamics. Across all values of p and q , when $\mathbb{E}[Y_i]$ is large, the index-based policy consistently achieves near-optimal performance. These results validate the theoretical insights regarding the effectiveness of the index structure under long-delay regimes.

In short, while the optimal policy offers the best theoretical performance, the index-based policy provides a near-optimal alternative with much lower computational cost, making it more suitable for real-time or resource-constrained systems.

VII. CONCLUSION

This article has proposed a sampling framework for remote monitoring systems based on the UoI, a state-dependent metric that captures the timeliness and relevance of updates. We first developed an optimal policy by a three-step *bisec-RVI* algorithm, while the suboptimal index-based policy offers a closed-form, low-complexity solution. Extensive numerical results have demonstrated the superiority of both proposed policies over existing baselines, with the index-based policy approaching optimality under relatively large delays. For future work, it would be interesting to analytically characterize the delay regimes under which the suboptimal policy becomes exactly optimal. In addition, extending the framework to more practical scenarios—specifically with parameter uncertainty and grant-free access—is a promising direction. This entails designing robust policies via reinforcement learning for unknown source statistics and enabling autonomous device-side transmission based on local uncertainty estimates.

APPENDIX A PROOF OF COROLLARY 2

Recall that the parameters p , and q satisfy $0 < p \leq q < 1$ and $p + q \neq 1$.

A. Lower Bound: $p^{(n)} > 0$

From the definition

$$p^{(n)} \triangleq \frac{p - p(1-p-q)^n}{p+q}$$

it suffices to show

$$(1-p-q)^n < 1.$$

Since $0 < p+q < 1$ or $1 < p+q < 2$, we have $|1-p-q| < 1$, and the inequality holds. Hence, $p^{(n)} > 0$.

B. Upper Bound: $p^{(n)} < 1$

We need to show

$$q + p(1-p-q)^n > 0.$$

This is straightforward for $0 < p+q < 1$. For $1 < p+q < 2$: 1) if $q > p$, then $q > -p(1-p-q)^n$, so the inequality holds and 2) if $q = p$, then

$$q + p(1-p-q)^n = p + p(1-2p)^n > 0$$

since $0 < p < 1$.

C. Range of $1 - q^{(n)}$

Since $q^{(n)}$ is symmetric to $p^{(n)}$, we can similarly show $0 < q^{(n)} < 1$ when $q \leq p$. For the case $1 < p+q < 2$ and $q \geq p$, we prove

$$1 - q^{(n)} = \frac{p + q(1-p-q)^n}{p+q} < 1.$$

Note that

$$\begin{aligned} p + q(1-p-q)^n &\geq p + q(1-p-q) \\ &= (1-q)(p+q) > 0. \end{aligned}$$

Hence, $0 < 1 - q^{(n)} < 1$, completing the proof.

APPENDIX B PROOF OF LEMMA 2

A. Proof of Part (i)

We prove that $\bar{p}_{\text{opt}} \leq \beta$ if and only if $S(\beta) \leq 0$.

1) $\bar{p}_{\text{opt}} \leq \beta$ if and only if $S(\beta) \leq 0$

If $\bar{p}_{\text{opt}} \leq \beta$, there exists a policy $\pi = (Z_1, Z_2, \dots) \in \Pi$ that is feasible for both (14a) and (16a) such that

$$\lim_{n \rightarrow \infty} \frac{\sum_{i=0}^{n-1} \mathbb{E}_{\pi} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} H(\tau^k(\omega)) \right]}{\sum_{i=0}^{n-1} \mathbb{E}_{\pi} [Z_{i+1} + Y_{i+1}]} \leq \beta \quad (46)$$

which is equivalent to

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_{\pi} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\tau^k(\omega)) - \beta] \right]}{\frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}_{\pi} [Z_{i+1} + Y_{i+1}]} \leq 0. \quad (47)$$

Since the intersampling times T_i are regenerative [45, Section VI.1], $Y_{i+1} > 0$, and $0 \leq Z_{i+1} < \infty$, the limit $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[Z_{i+1} + Y_{i+1}]$ always exists and positive, which means that

$$0 < \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E}[Z_{i+1} + Y_{i+1}] < \infty \quad (48)$$

for any policy. Thus, we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\tau^k(\omega)) - \beta] \right] \leq 0 \quad (49)$$

which implies that $S(\beta) \leq 0$, as $S(\beta)$ is the infimum value.

On the reverse direction, if $S(\beta) \leq 0$, we can know that there exists a policy $\pi = (Z_1, Z_2, \dots) \in \Pi$ that is feasible for both (14a) and (16a), satisfying (49). Because (48) always holds, we can obtain that (47) and (46) hold. Therefore, $\bar{p}_{\text{opt}} \leq \beta$.

2) $\bar{p}_{\text{opt}} > \beta$ if and only if $S(\beta) > 0$

If $\bar{p}_{\text{opt}} > \beta$, for any policy $\pi = (Z_1, Z_2, \dots) \in \Pi$ that is feasible for both (14a) and (16a), we have that

$$\lim_{n \rightarrow \infty} \frac{\sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} H(\tau^k(\omega)) \right]}{\sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1} + Y_{i+1}]} > \beta. \quad (50)$$

With (48) in hand, we derive that for any policy

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\tau^k(\omega)) - \beta] \right] > 0 \quad (51)$$

which implies that $S(\beta) > 0$.

On the contrary, if the infimum value $S(\beta) > 0$, then (51) holds for any policy. Similar to the proof of $\bar{p}_{\text{opt}} \leq \beta \Leftrightarrow S(\beta) \leq 0$, we can finally derive that $\bar{p}_{\text{opt}} > \beta$.

B. Proof of Part (ii)

From (16a), we can see that $S(\beta)$ is the pointwise infimum of the linear decreasing functions of β . Consequently, the root of $S(\beta) = 0$ is unique. Combining with the assertion in Part (i), we get that $\bar{p}_{\text{opt}}$ is the unique root of $S(\beta) = 0$.

C. Proof of Part (iii)

We show that any optimal solution to $\mathcal{P}4$ is also optimal for $\mathcal{P}3$, and vice versa. Since $S(\bar{p}_{\text{opt}}) = 0$ by Part (ii), there exists an optimal policy $\pi^* = (Z_1, Z_2, \dots)$ to $\mathcal{P}4$ such that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \left\{ \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} H(\tau^k(\omega)) \right] - \bar{p}_{\text{opt}} \mathbb{E} [Z_{i+1} + Y_{i+1}] \right\} = 0. \quad (52)$$

This implies that for policy π^*

$$\lim_{n \rightarrow \infty} \frac{\sum_{i=0}^{n-1} \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} H(\tau^k(\omega)) \right]}{\sum_{i=0}^{n-1} \mathbb{E} [Z_{i+1} + Y_{i+1}]} = \bar{p}_{\text{opt}} \quad (53)$$

which matches the objective of $\mathcal{P}3$. Hence, the policy π^* is optimal for both problems.

The converse follows by the same argument: any optimal policy for $\mathcal{P}3$ must also satisfy $S(\bar{p}_{\text{opt}}) = 0$ and, thus, be optimal for $\mathcal{P}4$. This completes the proof.

APPENDIX C PROOF OF LEMMA 1

For each fixed α , problem $\mathcal{P}5$ can be cast as a MDP problem with the same components as the CMDP in Section IV-B, except that $\bar{p}_{\text{opt}}$ in cost function is replaced by α .

Since the state and the action spaces are finite, an optimal unichain policy exists (see Appendix G for details). By dynamic programming [46, Proposition 7.4.1], the optimal policy satisfies Bellman's equations

$$\begin{aligned} s(\alpha) + V(\omega, \alpha) \\ = \inf_{Z_{i+1} \in \mathcal{N}} \left\{ C_i(Z_{i+1}, \omega; \alpha) + \mathbb{E} \left[\tau^{Z_{i+1}+Y_{i+1}}(\omega) V(1-q^{(Y_{i+1})}, \alpha) \right. \right. \\ \left. \left. + (1 - \tau^{Z_{i+1}+Y_{i+1}}(\omega)) V(p^{(Y_{i+1})}, \alpha) \right] \right\}, \text{ for all } \omega \in \mathcal{O} \end{aligned} \quad (54)$$

where $s(\alpha)$ is the optimal average cost and $V(\omega, \alpha)$ is the value function.

To solve this infinite-horizon MDP problem, we apply RVI [41, Section VI.5.2]. For numerical stability, we anchor the value function at state p by defining $\tilde{V}(\omega, \alpha) \triangleq V(\omega, \alpha) - V(p, \alpha)$, ensuring that $\tilde{V}(p, \alpha) = 0$. Then, (54) becomes

$$s(\alpha) + \tilde{V}(\omega, \alpha) = \inf_{Z_{i+1} \in \mathcal{Z}} \left\{ C_i(Z_{i+1}, \omega; \alpha) + r(Z_{i+1}, \omega; \alpha) \right\} \quad (55)$$

$$s(\alpha) = \inf_{Z_{i+1} \in \mathcal{Z}} \left\{ C_i(Z_{i+1}, p; \alpha) + r(Z_{i+1}, p; \alpha) \right\} \quad (56)$$

where

$$\begin{aligned} r(Z_{i+1}, \omega; \alpha) = \mathbb{E} \left[\tau^{Z_{i+1}+Y_{i+1}}(\omega) \tilde{V}(1-q^{(Y_{i+1})}, \alpha) \right. \\ \left. + (1 - \tau^{Z_{i+1}+Y_{i+1}}(\omega)) \tilde{V}(p^{(Y_{i+1})}, \alpha) \right]. \end{aligned} \quad (57)$$

Note that the anchoring state p is assumed to lie in the state space. If not, any other state can be chosen as the reference point. This completes the proof.

APPENDIX D DETAILS OF DISCUSSION 1

By definition, the optimal policy of the Lagrangian MDP satisfies (27b). To determine the optimal multiplier α^* , we examine the conditions in (27a), (27c), and (27d), which leads to two distinct cases.

1) *Inactive Constraint* ($\alpha^* = \bar{p}_{\text{opt}}$): (27c) implies that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n \mathbb{E}^{\pi_{\alpha^*}} [Z_{i+1} + Y_{i+1}] - \frac{1}{f_{\max}} \geq 0. \quad (58)$$

The frequency constraint is naturally satisfied, and the CMDP reduces to the unconstrained case. Hence, the optimal deterministic policy can be found by solving for the root of $G(\beta)$, as outlined in Algorithm 1 of [1].

2) *Active Constraint* ($\alpha^* > \bar{p}_{\text{opt}}$): In this case, (27d) implies that the constraint holds with equality

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^n \mathbb{E}^{\pi_{\alpha^*}} [Z_{i+1} + Y_{i+1}] - \frac{1}{f_{\max}} = 0. \quad (59)$$

The optimal solution is then a randomized mixture of two deterministic Lagrangian MDP policies, as described in [41, Theorem 6.6.2]. In this case, the optimal α^* can be found by solving for the root of (59).

APPENDIX E PROOF OF LEMMA 4

A. Proof of Part (i)

The Lagrangian dual function is typically expressed as

$$\begin{aligned} L(\pi_{\lambda}; \lambda) = \frac{\lambda}{f_{\max}} + \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{n=0}^{n-1} \mathbb{E}^{\pi_{\lambda}} \\ \times \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\Omega(R_i+k)) - \bar{p}_{\text{opt}} - \lambda] \right]. \end{aligned} \quad (60)$$

To align with our notation, we substitute $\lambda = \alpha - \bar{p}_{\text{opt}}$ and rewrite

$$L(\pi_{\alpha}; \alpha) = \Phi(\pi_{\alpha}; \alpha) + \frac{\alpha - \bar{p}_{\text{opt}}}{f_{\max}}. \quad (61)$$

According to [42, Lemma 3.1], for $\alpha > \bar{p}_{\text{opt}}$ ($\lambda > 0$) and any $\xi > 0$, the following inequalities hold:

$$\begin{aligned} L(\pi_{\alpha}; \alpha + \xi) - L(\pi_{\alpha}; \alpha) &\geq L(\pi_{\alpha+\xi}; \alpha + \xi) - L(\pi_{\alpha}; \alpha) \\ &\geq L(\pi_{\alpha+\xi}; \alpha + \xi) - L(\pi_{\alpha+\xi}; \alpha). \end{aligned} \quad (62)$$

This leads to

$$-\xi h(\pi_{\alpha}) \geq L(\pi_{\alpha+\xi}; \alpha + \xi) - L(\pi_{\alpha}; \alpha) \geq -\xi h(\pi_{\alpha+\xi}) \quad (63)$$

implying that $h(\pi_{\alpha}) \leq h(\pi_{\alpha+\xi})$; thus, we finish the proof.

B. Proof of Part (ii)

We first show that $\lim_{\alpha \rightarrow 0} h(\pi_{\alpha}) \leq 0$. Because the optimal policy satisfies $h(\pi_{\alpha^*}) = 0$, $\alpha > 0$, and $h(\pi_{\alpha})$ is nondecreasing in α [from Part (i)], we get that

$$\lim_{\alpha \rightarrow 0} h(\pi_{\alpha}) \leq 0. \quad (64)$$

Next, we show that $h(\pi_1) > 0$. For $\alpha \geq 1$, since $0 < H(x) < 1$, we have $H(x) - \alpha < 1 - \alpha = 0$ for all $x \in (0, 1)$, leading to

$$C_i(Z_{i+1}, \Omega(R_i); \alpha) < 0, \quad \forall i. \quad (65)$$

Thus, the optimal policy under this setting selects the maximum waiting time $Z_{i+1} = M$ for all i , yielding

$$h(\pi_{\alpha}) = M + \mathbb{E}[Y_{i+1}] - \frac{1}{f_{\max}} > 0. \quad (66)$$

This completes the proof.

APPENDIX F PROOF OF LEMMA 6

When the constraint is inactive, the result $\bar{p}_{\text{opt}} = \alpha^*$ follows directly from Discussion 1.

For the case where the constraint is active, we have $\bar{p}_{\text{opt}} < \alpha^*$. Under strong duality and from (26), the optimal pair $(\pi_{\alpha^*}^*, \alpha^*) = (\pi_{\bar{p}_{\text{opt}}}^*, \bar{p}_{\text{opt}})$ satisfies

$$S(\bar{p}_{\text{opt}}) = s(\alpha^*) + \frac{\alpha^* - \bar{p}_{\text{opt}}}{f_{\max}}. \quad (67)$$

Since $S(\bar{p}_{\text{opt}}) = 0$, we rearrange (67) to obtain

$$\bar{p}_{\text{opt}} = f_{\max} s(\alpha^*) + \alpha^*. \quad (68)$$

This completes the proof.

APPENDIX G PROOF OF THEOREM 2

To establish the convergence of the RVI algorithm for $\mathcal{P}5$, we rely on the following sufficient conditions from [47, Proposition 4.3.2].

Lemma 10: If an MDP satisfies the following conditions: 1) the MDP is a unichain MDP and 2) the optimal stationary policy for the MDP induces an aperiodic transition probability matrix, then the sequences $\{s_k(\alpha)\}_{k \in \mathbb{N}}$ and $\{\tilde{V}_k(\omega, \alpha)\}_{k \in \mathbb{N}}$ converge to the solution of Bellman's optimality equation (54)

$$\begin{aligned} \lim_{k \rightarrow \infty} s_k(\alpha) &= s(\alpha) \\ \lim_{k \rightarrow \infty} \tilde{V}_k(\omega, \alpha) &= \tilde{V}(\omega, \alpha), \text{ for all } \omega \in \mathcal{O}. \end{aligned} \quad (69)$$

To prove convergence, we verify both conditions.

A. Unichain Property

From (17), the belief state transitions as $\tau^{Z_{i+1}+Y_{i+1}}(\Omega(R_i)) \in \{p^{(n)}, 1 - q^{(n)}\}_{n \in \mathbb{N}}$. By Corollary 2, we know that, for all n , $0 < p^{(n)} < 1$ and $0 < 1 - q^{(n)} < 1$, implying that

$$P[\Omega(R_{i+1}) | \Omega(R_i)] > 0, \quad \forall \Omega(R_i), Z_{i+1}.$$

Therefore, every state can reach every other state (including itself) with positive probability, confirming that the MDP is unichain [40, Section 8.3].

B. Aperiodicity

Because all transition probabilities are strictly positive, each state has a nonzero probability of remaining in itself. Hence, the induced transition matrix is aperiodic.

As both conditions are satisfied, the convergence of the RVI algorithm is guaranteed.

APPENDIX H PROOF OF THEOREM 3

For each fixed ρ , the problem $\mathcal{P}8$ can be cast as an average-cost MDP, with the same components as the CMDP in Section V-B, except that the cost function replaces $\bar{p}_{\text{nopt}}$ with ρ .

This MDP $\mathcal{P}8$ can be solved by using dynamic programming [46, Proposition 7.4.1]. Given that $\Omega(R_i) = \omega$, the

optimal waiting time Z_{i+1} for the $(i+1)$ th packet satisfies Bellman's optimality equation

$$\begin{aligned} s_1(\rho) + V_1(\omega, \rho) &= \inf_{Z_{i+1} \in \mathbb{N}} \left\{ \mathbb{E} \left[\sum_{k=0}^{Z_{i+1}+Y_{i+1}-1} [H(\tau^k(\omega)) - \rho] \right] \right\} \\ &\quad + \mathbb{E} \left[\omega^* V_1(1 - q^{(Y_{i+1})}, \rho) + (1 - \omega^*) V_1(p^{(Y_{i+1})}, \rho) \right] \end{aligned} \quad (70)$$

for all $\omega \in \mathcal{O}$, where $s_1(\rho)$ is the average cost and $V_1(\omega)$ is the relative value function.

Let $\bar{Z} = Z(\omega, \rho)$ be the optimal solution to (70), which means that, given that $\Omega(R_i) = \omega$, then the optimal waiting time Z_{i+1} for sending the $(i+1)$ th packet is $\bar{Z}$. Since the second expectation term is independent of $\bar{Z}$, minimizing the cost reduces to

$$\inf_{\bar{Z} \in \mathbb{N}} \mathbb{E} \left[\sum_{k=0}^{\bar{Z}+Y_{i+1}-1} (H(\tau^k(\omega)) - \rho) \right]. \quad (71)$$

We now derive a condition under which $\bar{Z} = 0$ is optimal. This occurs if

$$\begin{aligned} \inf_{\bar{Z} \in \mathbb{N}^+} \mathbb{E} \left[\sum_{k=0}^{\bar{Z}+Y_{i+1}-1} (H(\tau^k(\omega)) - \rho) \right] \\ \geq \mathbb{E} \left[\sum_{k=0}^{Y_{i+1}-1} (H(\tau^k(\omega)) - \rho) \right]. \end{aligned} \quad (72)$$

Rearranging terms yields

$$\inf_{\bar{Z} \in \mathbb{N}^+} \mathbb{E} \left[\sum_{k=0}^{\bar{Z}-1} [H(\tau^{k+Y_{i+1}}(\omega)) - \rho] \right] \geq 0. \quad (73)$$

Because the left-hand side of (73) is the infimum of a set of strictly increasing linear functions in ρ , (73) holds if and only if

$$\inf_{\bar{Z} \in \mathbb{N}^+} \left\{ \frac{1}{\bar{Z}} \mathbb{E} \left[\sum_{k=0}^{\bar{Z}-1} [H(\tau^{k+Y_{i+1}}(\omega))] \right] \right\} \geq \rho. \quad (74)$$

The left-hand side of (74) corresponds to the index function $\eta(\omega)$ defined in (39).

Similarly, $\bar{Z} = 1$ is optimal if $\bar{Z} \neq 0$ and

$$\inf_{\bar{Z} \in \{2, 3, \dots\}} \mathbb{E} \left[\sum_{k=0}^{\bar{Z}-1} [H(\tau^{k+Y_{i+1}}(\omega)) - \rho] \right] \geq 0 \quad (75)$$

which is equivalent to

$$\eta(\tau(\omega)) \geq \rho. \quad (76)$$

By repetition of the same steps as in (74)–(75), we conclude that $\bar{Z} = k$ is optimal if $\eta(\tau^k(\omega)) \geq \rho$ and $\bar{Z} \neq 0, 1, \dots, k-1$. Thus, the optimal waiting time $Z_{i+1} = \bar{Z}$ satisfies

$$\bar{Z} = Z(\omega, \rho) = \min \{k \in \mathbb{N} : \eta(\tau^k(\omega)) \geq \rho\}. \quad (77)$$

This completes the proof.

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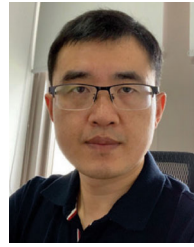


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